

SOFTWARE ENGINEERING LAB – 2

(UE24CS341A)

Agile Backlog Creation & Sprint Simulation in Jira

Team Number: 003

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Problem Statement 3: Campus Placement & Internship Pipeline

EPIC: Epic 2 – Drive Configuration & Peak Performance (FR-002 and NFR-002)

Note: This report covers only Epic 2, my individual contribution from the Lab 1 report. The remaining requirements are covered separately by my teammates (Epic 1 – FR-001: Mounik; Epic 3 – FR-003 and FR-005: Mallikarjuna; Epic 4 – FR-004 and NFR-001: Moksha).

Lab Objective

This lab converts the requirements FR-002 and NFR-002 from the Lab 1 report into an Agile Epic and User Stories in Jira, estimates effort using the Fibonacci scale, prioritises the backlog, simulates two one-week sprints on a Scrum board, and tracks sprint progress using the Burndown Chart.

1. Epic (Derived from FR-002 and NFR-002)

The two requirements are grouped into one Epic: FR-002 defines how eligibility rules are configured per drive, and NFR-002 defines the load the platform must handle when students apply against those drives.

Epic Key	Epic Name	Mapped Requirements	Description
PPK-1	Epic 2: Drive Configuration & Peak Performance	FR-002, NFR-002	Enable Placement Officers to define flexible CGPA and backlog eligibility rules for each company drive, and ensure the platform handles peak application load (500 concurrent students) within 3 seconds.

2. User Stories for Epic 2

The Epic is broken into 6 User Stories using the format: As a [role], I want [goal], so that [benefit].

Key	Story	As a...	I want...	So that...	Covers
PPK-2	2.1 Set Minimum CGPA per Drive	Placement Officer	to set a minimum CGPA for each company drive	only academically eligible students can apply	FR-002
PPK-3	2.2 Set Maximum Active Backlogs	Placement Officer	to set the maximum allowed active backlogs for a drive	students exceeding the limit are filtered out automatically	FR-002
PPK-4	2.3 Multi-Tier Eligibility Rules	Placement Officer	to define different CGPA and backlog tiers for one company (e.g. Dream vs Regular offer)	eligibility matches the company's hiring structure	FR-002
PPK-5	2.4 Preview Eligible Applicant List	Placement Officer	to preview the students who meet a drive's criteria before publishing it	I can verify the filters work correctly	FR-002

Key	Story	As a...	I want...	So that...	Covers
PPK-6	2.5 Fast Applications Under Peak Load	Student Applicant	application pages to respond within 3 seconds even when hundreds of students apply at once	I can submit before the deadline without lag	NFR-002
PPK-7	2.6 Peak Load Testing (500 Users)	Placement Officer	the platform verified under 500 concurrent users	I can trust it during placement season	NFR-002

Acceptance criteria were taken from Lab 1. For example, FR-002: *a drive configured with CGPA ≥ 7.5 and 0 backlogs excludes all applicants below the threshold*; NFR-002: *load testing with 500 concurrent users shows a 95th-percentile response time of 3 seconds or less*.

3. Jira Backlog – Epic, User Stories and Story Points

The screenshot shows Epic 2 with all six User Stories, the priority icon and story points on each row, and the Epic panel reporting 6 work items and an estimate of 29 points.

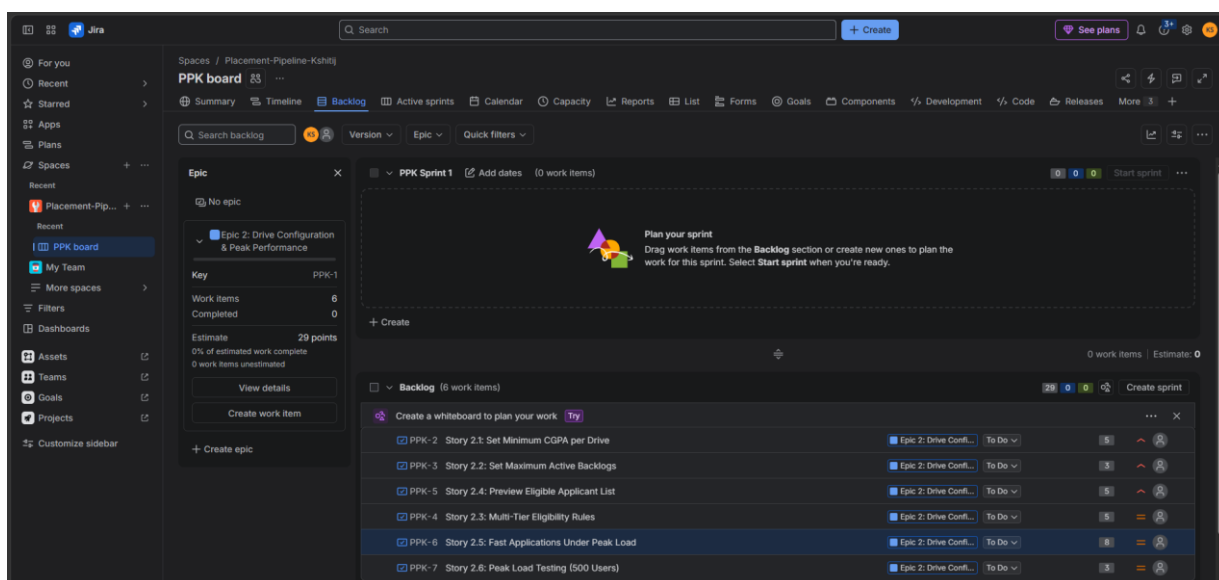


Figure 1: Jira Backlog – Epic 2 (PPK-1) with stories PPK-2 to PPK-7 (5, 3, 5, 5, 8, 3 pts = 29 pts).

4. Backlog Prioritization

User Stories were ordered by business value and dependency. The backlog order in Jira is 2.1, 2.2, 2.4, 2.3, 2.5, 2.6.

Key	Story	Priority	Story Points
PPK-2	2.1 Set Minimum CGPA per Drive	High	5
PPK-3	2.2 Set Maximum Active Backlogs	High	3
PPK-5	2.4 Preview Eligible Applicant List	High	5
PPK-4	2.3 Multi-Tier Eligibility Rules	Medium	5
PPK-6	2.5 Fast Applications Under Peak Load	Medium	8
PPK-7	2.6 Peak Load Testing (500 Users)	Medium	3

- **2.1 and 2.2 (High):** the basic CGPA and backlog rules are the core of FR-002; every other story depends on them.

- **2.4 (High):** the preview lets the officer verify that the rules work before a drive is published, so it follows directly after the rules.
- **2.3 (Medium):** multi-tier rules extend the basic rules and are only useful once those work.
- **2.5 and 2.6 (Medium):** performance and load testing make sense once there is functionality to measure.

Total Story Points (Epic 2): 29 | High: 13 | Medium: 16

5. Story Point Estimation – Planning Poker

Story points were estimated independently by each team member using the Fibonacci scale (1, 2, 3, 5, 8, 13...). The individual estimates were compared, differences were discussed, and the team agreed on the final estimates.

Story ID	Story Title	Kshitij	Mallikarjuna	Mounik	Moksha	Agreed
PPK-2 / 2.1	Set Minimum CGPA per Drive	5	5	5	3	5
PPK-3 / 2.2	Set Maximum Active Backlogs	3	3	3	3	3
PPK-4 / 2.3	Multi-Tier Eligibility Rules	5	5	5	5	5
PPK-5 / 2.4	Preview Eligible Applicant List	5	5	5	3	5
PPK-6 / 2.5	Fast Applications Under Peak Load	8	8	8	8	8
PPK-7 / 2.6	Peak Load Testing (500 Users)	3	3	3	3	3

6. Sprint Planning & Simulation

Sprint 1 (Duration: 1 week)

Sprint Goal: Placement Officers can configure and verify basic drive eligibility rules.

Key	Story	Priority	Pts	Status
PPK-2	2.1 Set Minimum CGPA per Drive	High	5	Done
PPK-3	2.2 Set Maximum Active Backlogs	High	3	Done
PPK-5	2.4 Preview Eligible Applicant List	High	5	Done

Sprint 1 Velocity: 13 pts committed | 13 pts completed | 0 pts rolled over

Sprint 2 (Duration: 1 week)

Sprint Goal: Support multi-tier rules and ensure the platform handles peak load.

Key	Story	Priority	Pts	Status
PPK-4	2.3 Multi-Tier Eligibility Rules	Medium	5	Done
PPK-6	2.5 Fast Applications Under Peak Load	Medium	8	Done
PPK-7	2.6 Peak Load Testing (500 Users)	Medium	3	Done

Sprint 2 Velocity: 16 pts committed | 16 pts completed | 0 pts rolled over

Jira Sprint Board (Active Sprint view) – Sprint 1

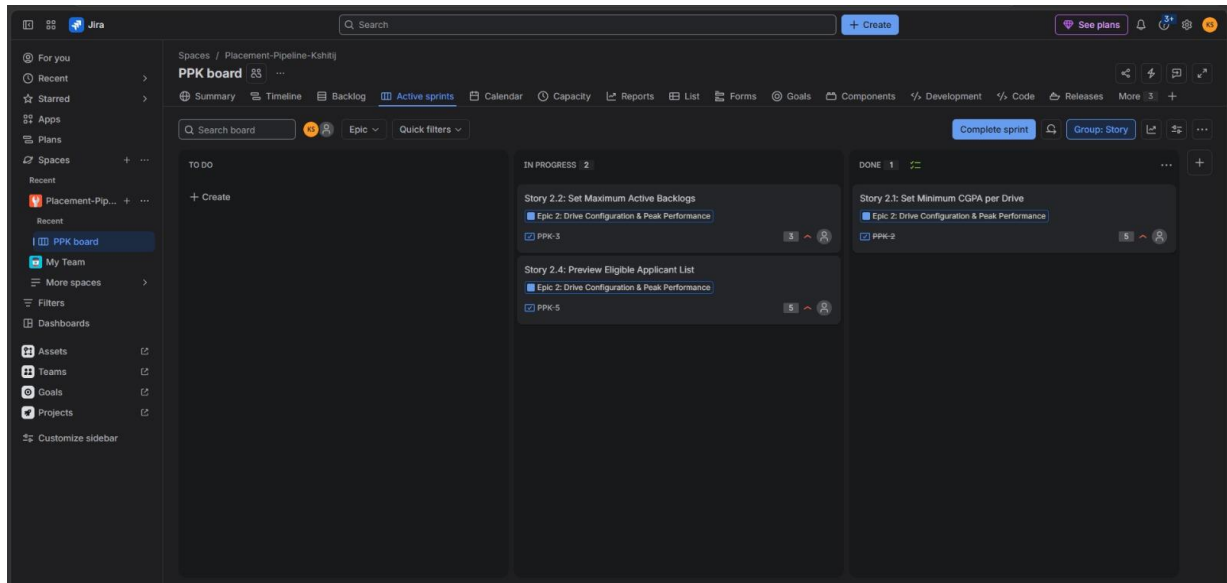


Figure 2: Sprint 1 Active Sprint board mid-sprint – Story 2.1 in DONE; Stories 2.2 and 2.4 IN PROGRESS.

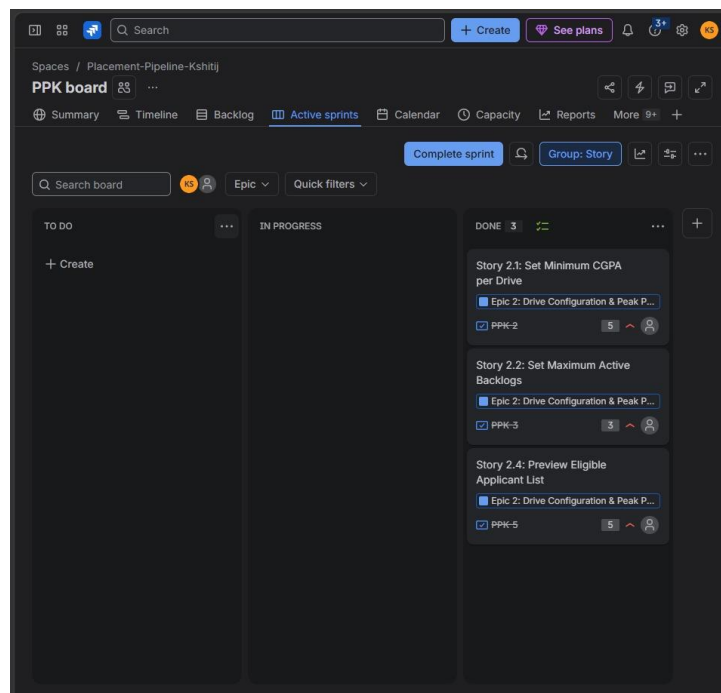


Figure 3: Sprint 1 Active Sprint board – all three stories (13 pts) in DONE, before clicking Complete sprint.

Sprint 2 outcome

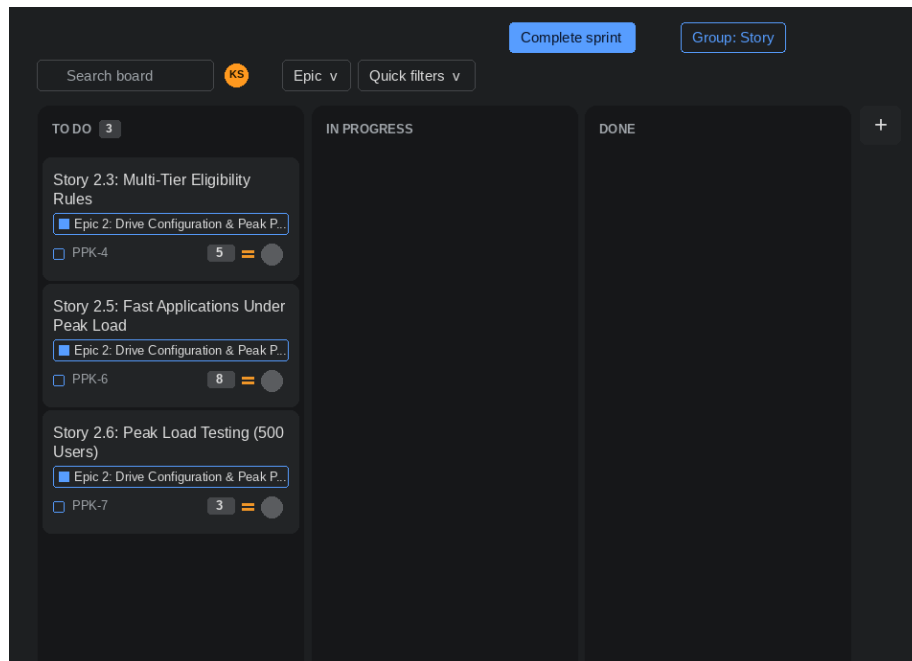


Figure 4: Sprint 2 Active Sprint board – all three stories (13 pts) in TO DO.

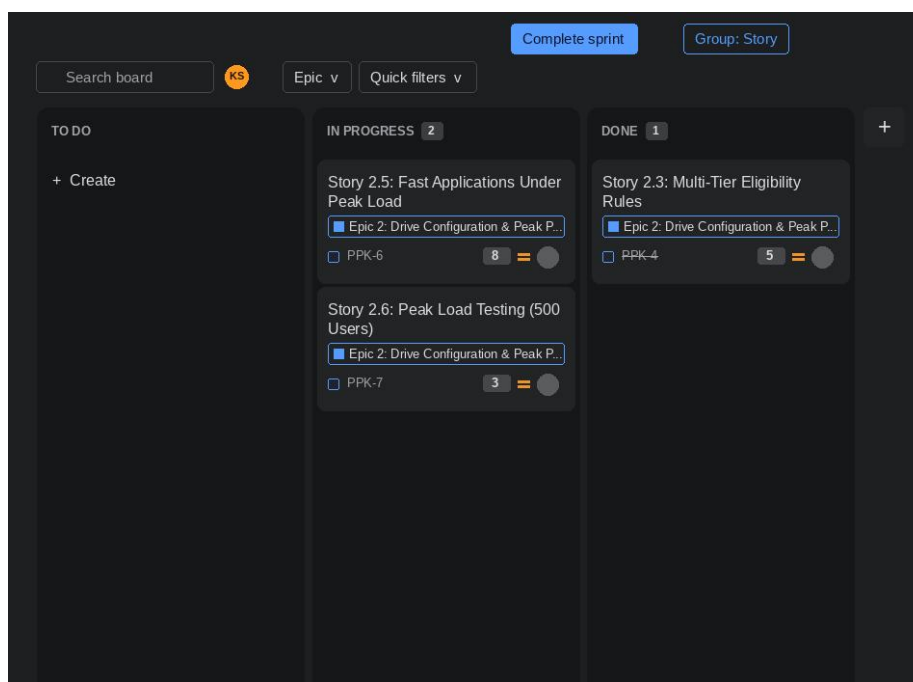


Figure 5: Sprint 2 Active Sprint board mid-sprint – Story 2.3 in DONE; Stories 2.5 and 2.6 IN PROGRESS.

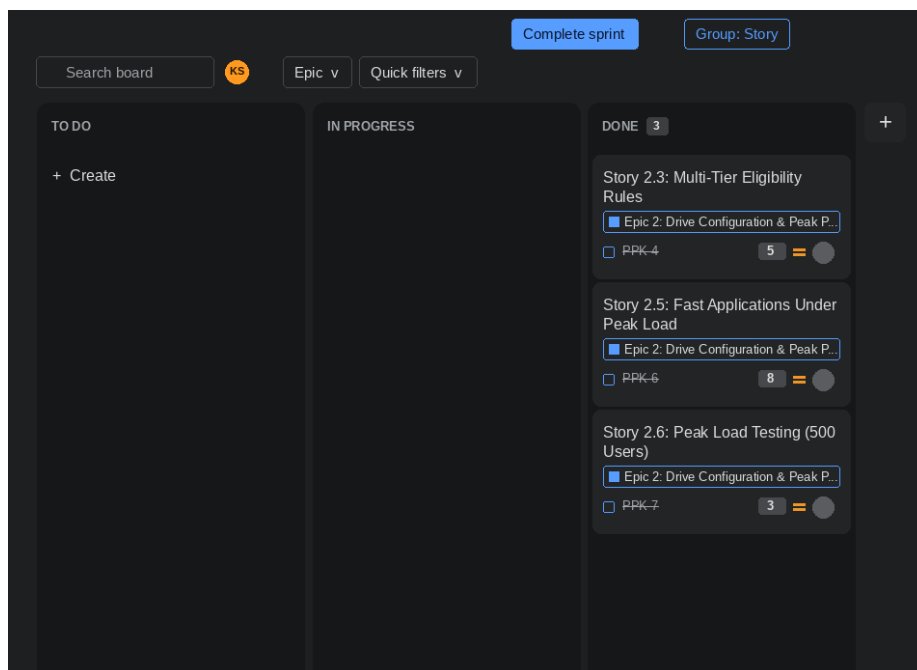


Figure 6: Sprint 2 Active Sprint board – all three stories (16 pts) in DONE, before clicking Complete sprint.

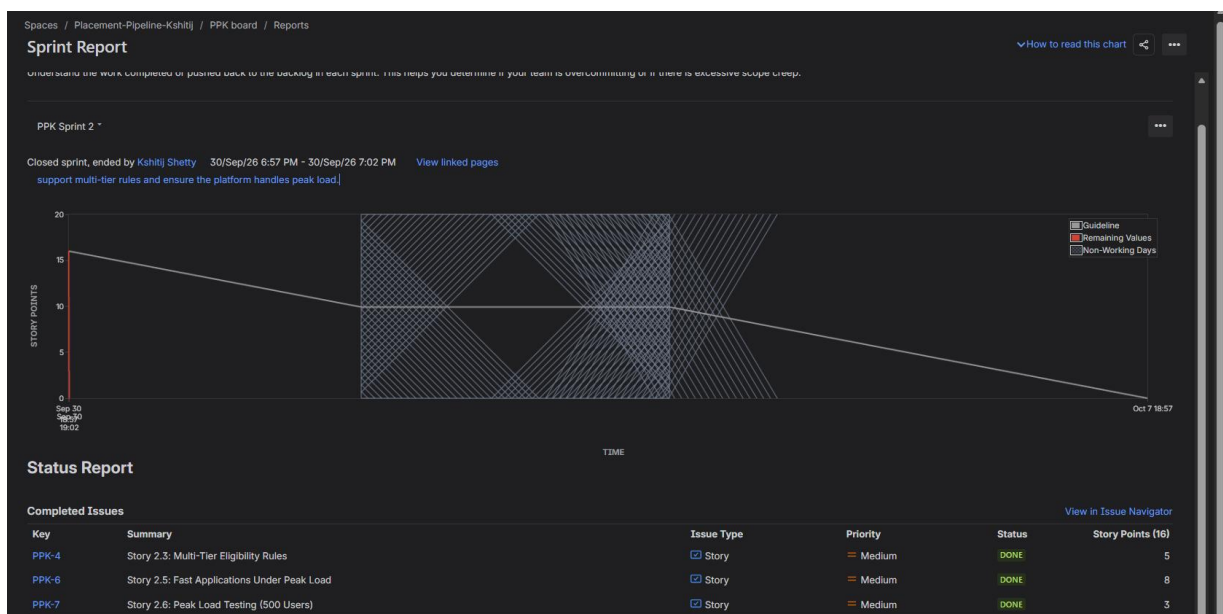


Figure 7: Sprint 2 outcome (Sprint Report, taken after completion) – PPK-4, PPK-6 and PPK-7 DONE, 16 story points.

7. Burndown Chart Analysis

The Jira Burndown Chart plots story points remaining (red line) against the ideal guideline (grey line). Each sprint is shown twice: once with non-working days displayed (hatched) and once with them hidden, which makes the guideline easier to read. The event table under each chart lists the exact completion times.

Sprint 1 Burndown (13 pts)

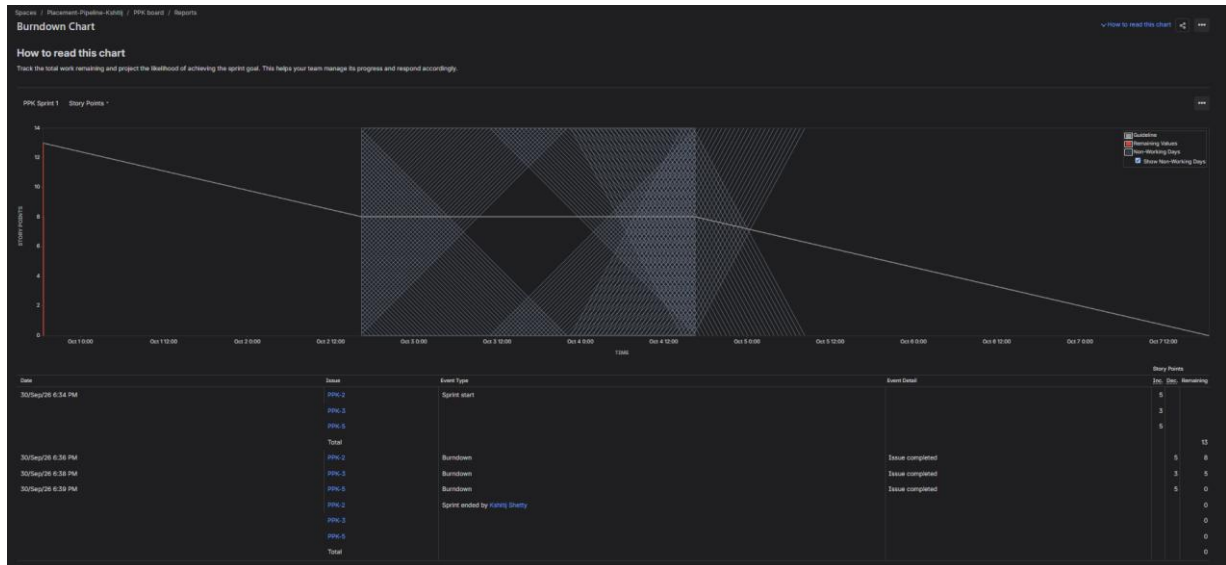


Figure 8: Sprint 1 Burndown, non-working days shown. $13 \rightarrow 8 \rightarrow 5 \rightarrow 0$ between 6:36 PM and 6:39 PM on 30 Sep.

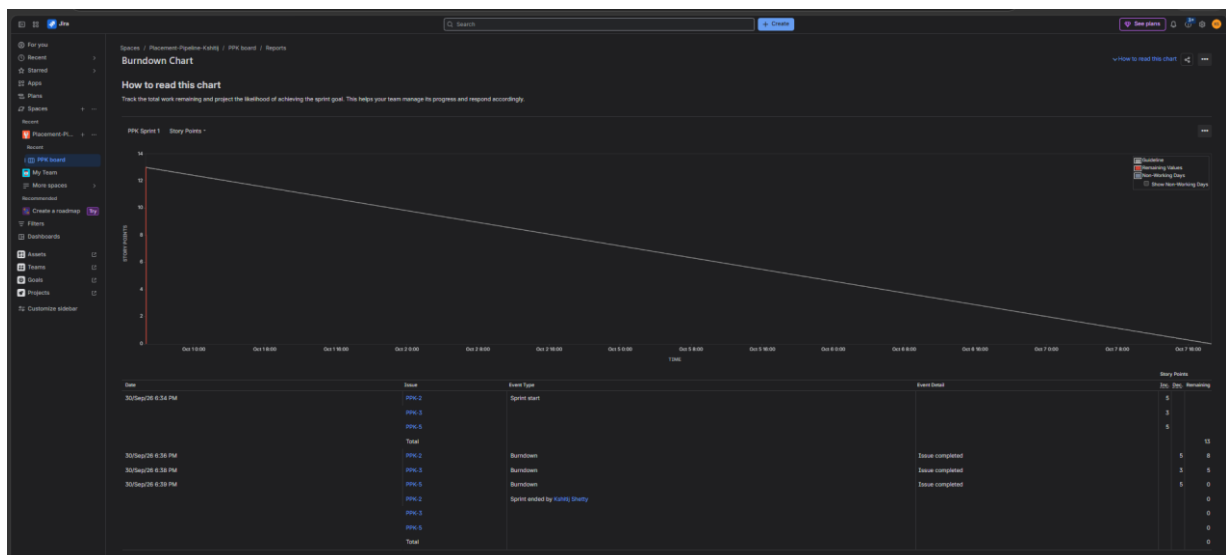


Figure 9: Sprint 1 Burndown, non-working days hidden.

Sprint 2 Burndown (16 pts)

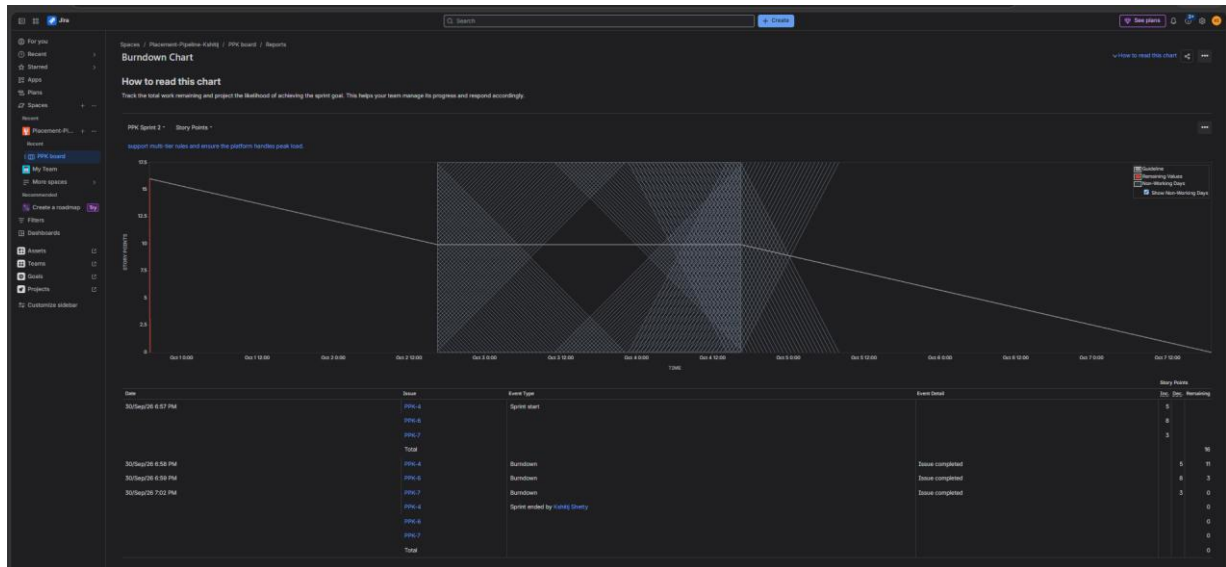


Figure 10: Sprint 2 Burndown, non-working days shown. 16 → 11 → 3 → 0 between 6:58 PM and 7:02 PM on 30 Sep.

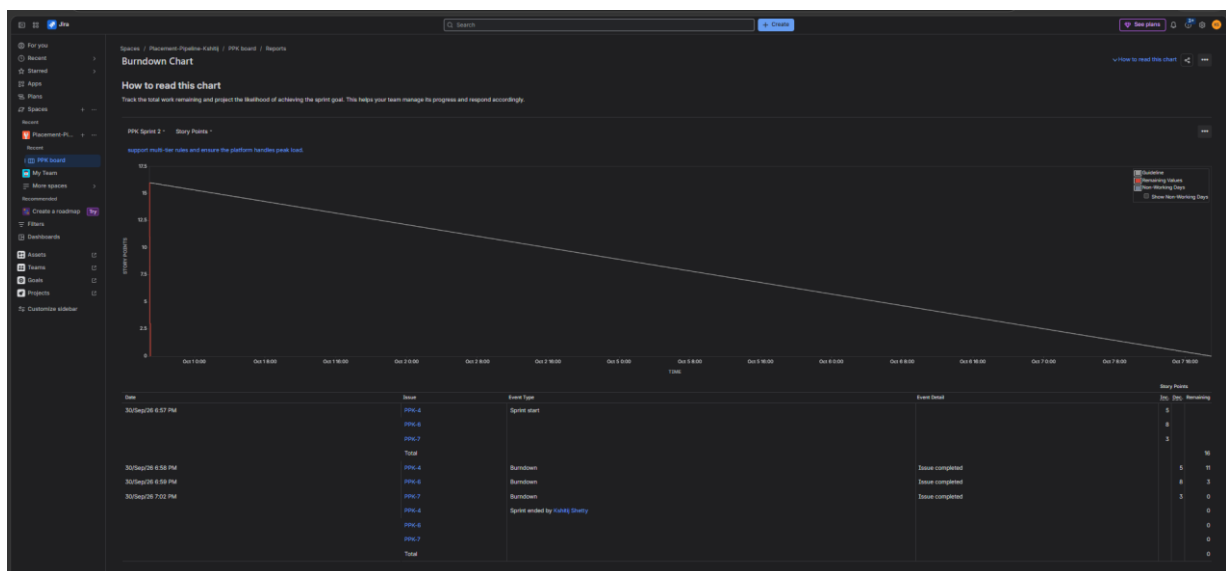


Figure 11: Sprint 2 Burndown, non-working days hidden.

Summary of the burndown data (from the Jira event tables)

Sprint	Start (points)	Completion events	End (points)
Sprint 1	6:34 PM – 13	6:36 PM PPK-2 (-5) → 8; 6:38 PM PPK-3 (-3) → 5; 6:39 PM PPK-5 (-5) → 0	0
Sprint 2	6:57 PM – 16	6:58 PM PPK-4 (-5) → 11; 6:59 PM PPK-6 (-8) → 3; 7:02 PM PPK-7 (-3) → 0	0

8. Reflection Questions

1. Did your estimations reflect the actual effort?

We can only partly judge this. The sprints were simulated by moving cards on the board, so no real development effort was spent and the estimates cannot be checked against actual hours. What the exercise did show is that the relative sizing was sensible: the load-related story 2.5 received the highest estimate (8) because performance work carries the most uncertainty, while the rules 2.1 and 2.2 were sized relative to each other (5 and 3). In a real sprint we would expect 2.5 to be the one most likely to

overrun. The individual estimates were compared, differences were discussed, and the team agreed on the final estimates. This helped align the estimates with the relative complexity and uncertainty of each story.

2. Was your backlog well-prioritized?

Yes. The three High-priority stories (2.1, 2.2, 2.4) form the core of FR-002 and were placed in Sprint 1, so that eligibility rules could be configured and verified first. Story 2.3 builds on the basic rules, and stories 2.5 and 2.6 measure performance, which only makes sense once the functionality exists. One possible improvement: load testing (2.6) could begin earlier on a partial build so that performance problems are found before the final sprint.

3. How did your simulated sprint align with your plan?

Both sprints were completed as planned. Sprint 1 committed and completed 13 points and Sprint 2 committed and completed 16 points, a total of 29 points with nothing rolled over, and both sprint goals were met. The board moved stories from To Do to In Progress to Done in order of priority. Because the work was simulated, each sprint was finished within minutes instead of the one week that was planned, so the 100% completion reflects the simulation, not real delivery capacity.

4. What insights did the burndown chart give about your team's capacity?

In both sprints the red line dropped to zero within minutes of the start, far below the grey guideline, which is why it appears as a thin line at the left edge of the chart. In a real sprint that pattern would indicate that the team had under-committed. The event tables still gave useful information: Sprint 1 burned down 13 points and Sprint 2 burned down 16 points, so a realistic velocity would be somewhere around 13–16 points per week, but this would need to be re-measured with real work. The charts also showed that 2.5, the 8-point story, accounted for half of Sprint 2's burndown in one step, which suggests large stories should be split so that progress is visible throughout the sprint.